Warren Kozak

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Education

Carleton College — September 2023 – June 2027
Bachelor of Arts in Computer Science, Mathematics; Cumulative GPA: 3.9/4.0 — *Northfield, MN*

- Relevant Coursework: Data Structures, Algorithms, Advanced Software Design, Programming Languages, Introduction to Computer Systems, Robotics, Multivariable Calculus, Linear Algebra, Ordinary Differential Equations, Computational Linear Algebra

Research Experience

Robotics Researcher (Coursework) — April – June 2026
Carleton College — *Northfield, MN*

- TBD

NSF REU Undergraduate Researcher — July 2025 – August 2025
Northwestern University (TIILT Lab) — *Evanston, IL*

- Prototyped wearable devices for use in swimming using a Seeed Studio nRF52840 Sense
- Optimized data logging with FreeRTOS, increasing IMU sample rate from 25 to 50 Hz for more reliable data
- Utilized a Mahony filter algorithm to generate quaternions from 6-axis IMU data to detect device orientation
- Developed an LSTM model to classify IMU data into stroke categories, achieving an accuracy of 95%

Industry Experience

Garmin - GNSS Platforms Team — June 2026 – August 2026
Embedded Software Engineering Intern — *Olathe, KS*

- TBD - Worked with GNSS team

Hildebrand Technology — June 2024 – August 2024
Summer Intern — *London, UK*

- Developed a Home Assistant integration in Python supporting Hildebrand Glow IoT devices via MQTT
- Integrated Glow data stream with the UK National Grid API to deliver actionable insights to users
- Authored comprehensive documentation to guide users through the integration setup process

Projects

N-Body Physics Sim | *C++, Raylib, AGAMA* — March 2026 – Present

- TBD

Scheme Interpreter | *C, Scheme* — April – June 2025

- Implemented a 1000-line Guile Scheme interpreter in C consisting of a tokenizer, parser, and interpreter, supporting recursion, closures, special forms (if, let, define), and primitives (map, cons, car, cdr)
- Developed a custom implementation of talloc, achieving 0 memory leaks under Valgrind

Activities

Men's Varsity Swim & Dive | *Carleton College* — September 2023 – Present

- Elected by teammates to serve as team captain for the 2026–2027 season
- Swam at 2025 MIAC Championships, achieving all-MIAC in the 200 medley and freestyle relays
- CSC Academic All-District (2025–2026)

Technical Skills

Languages: C++, C, Python, Lisp/Scheme, LaTeX
Frameworks: ROS2, FreeRTOS, Arduino, Flask
Developer Tools: UNIX/Linux, Git, PlatformIO, Github, Docker, Neovim
Libraries: Keras, TensorFlow, pandas, NumPy, SciPy, OpenCV, NLTK, Matplotlib